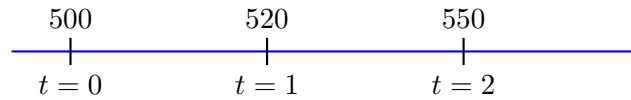


Problem 3.5

An initial deposit of 500 accumulates to 520 at the end of one year and 550 at the end of the second year. Find i_1 and i_2 .

Solution

The timeline is:



For the first year,

$$i_1 = \frac{A(1) - A(0)}{A(0)}.$$

Thus,

$$i_1 = \frac{520 - 500}{500} = \frac{20}{500} = 0.04 = 4\%.$$

For the second year,

$$i_2 = \frac{A(2) - A(1)}{A(1)}.$$

Thus,

$$i_2 = \frac{550 - 520}{520} = \frac{30}{520} \approx 0.05769 = 5.77\%.$$

Therefore,

$$\boxed{i_1 = 4\%, \quad i_2 \approx 5.77\%.$$